

LINGKANG JIN, PHD

| Energy systems modeler | ML Data scientist | R&D engineer

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PhD Energy Systems Modeler & ML Specialist based in Eindhoven. **Work Package Leader** for national research projects, coordinating researchers and industrial partners including grid operators like **Enexis, Stedin, and DNV**. Developed AI-driven surrogates achieving **20x computational speedups for grid optimization**. Expert in optimization domain knowledge with scalable data science to drive the Dutch energy transition.

SKILLS

Optimization	• Mathematical Programming: LP, MILP, SOCP, MIQP, Multi-objective, and Data-driven optimization • Solvers & Frameworks: <i>Pyomo, Gurobi, MOSEK, HiGHS, CVXpy, Gurobipy, Calliope</i> • Spatial Modeling: Geo-referenced optimization for grid infrastructure
Machine Learning	• Surrogate Modeling: AI-driven optimization using Graph Neural Networks (GNN) and Random Forests • Forecasting: Time series forecasting (ARIMA-Statistical hybrids) for load and PV production • Toolkit: <i>PyTorch, MLflow, Numpy, Pandas, Scikit-learn, Seaborn, Plotly</i>.
Engineering	• Energy Systems: Planning, Grid Power Flow, Congestion Management, Energy Community Design • Thermal & Mechanical: CFD, Heat Exchanger design, CAD, and Cooling solution architectures
Software Eng.	• DevOps & Tools: Git, Gitlab, CI/CD pipelines, Sphinx, Read the Docs. • Deployment: REST API development, Flask, Streamlit, Dash, and High-Performance Computing (HPC)

EXPERIENCE

Jan 2024 – Now	Postdoctoral Researcher <i>Eindhoven University of Technology, NL</i> • Design, deployment and continuous improvement of the autonomous cloud computing optimization model using LP, MILP and MIQP approach for the day ahead assets (home batteries) steering for the end-users, achieving 5-6% grid relief in a demonstration project on low voltage grid in the Netherlands With Enexis, CGI and iWell. • Develop the rolling horizon load and PV production time series forecast, based on the weather dependency and an hybrid ARIMA and statistical approach. • Accelerated DSO-Aggregator bilevel optimization by 10–20x via Graph Neural Network surrogate modeling, significantly reducing complexity compared to traditional MILP/NLP approaches • Responsible for the dedicated work packages in two national research projects, with the role to coordinating up to 4 researchers and support to the industrial partners as Enexis, Stedin, DNV
Oct 2020 – Nov 2023	Energy Systems Ph.D. Candidate <i>Università Politecnica delle Marche, IT</i> • Active contribution for a four-year Horizon Europe consortium (€5M+ budget) and two national projects; coordinated multi-partner deliverables, from initial literature reviews to full-scale algorithm deployment. • Design of the multi-objective (economic and environmental) energy planning of the multi-energy carrier community, applied in the context of the medium scale university campus with 15,000+ students. • Spearheaded the establishment of a Hydrogen Systems Laboratory, overseeing vendor procurement, regulatory compliance, and the development of technical fire hazard safety protocols. • Secured two international collaborations and visiting periods and joint research works, specifically with Technical University of Denmark and Eindhoven University of Technology.
Sep 2020 – Oct 2023	Engineering Consultant <i>Freelance, Remote</i> • Retained R&D Thermal Engineer for <i>Nrgeer</i>, leading thermal system design and project engineering through advanced CAD modeling and simulation. • Supply Chain & Sales Consultant for <i>CAP-XX</i>; optimized the procurement strategy for supercapacitors by identifying and qualifying new global suppliers, as well as clients expansion.
Dec 2019 – Sep 2020	Lab Engineer <i>Whirlpool EMEA, IT</i> • Validated prototype compliance with updated NVH (Noise, Vibration, and Harshness) regulatory standards

EDUCATION

Oct 2020 – Mar 2024	Ph.D. in Energy Systems <i>Università Politecnica delle Marche, IT</i> Thesis: <i>Energy storage in multi-energy carrier communities: Li-ion batteries and hydrogen multi-physical details for integration into the planning stage.</i>
Sep 2017 – Nov 2019	M.Sc. in Mech. Engineering <i>Università Politecnica delle Marche, IT</i> Thesis: <i>Wind resource assessment for a preliminary offshore wind farm in the Mediterranean sea.</i>
Sep 2014 – Jul 2017	B.Sc. in Mech. Engineering <i>Università Politecnica delle Marche, IT</i> Thesis: <i>Matlab code development for the Continuous Laser Doppler Vibrometry testing.</i>

ADDITIONAL INFORMATION

Teaching & Supervision • Delivered 12+ hours of university lectures. • Mentored 7 MSc candidates. • Speaker at 10 international conferences and published 10+ journal papers.
Languages • English (Fluent) • Italian (Native) • Chinese (Native) • Dutch (Elementary / In Progress)
Certifications • Full European Driving License (Cat. B)

ADDITIONAL INFORMATION

PROJECTS

ORKEST Optimal integration of network flexibility and asset intelligence to increase large-scale integration of RES, while maintaining reliability

Netherlands Enterprise Agency (RVO)

📅 2025-2026

📍 Netherlands

Congestion management and asset control integration for the RES integration with the active involvement of DSO (Stedin)

Consortium: Stedin, Technical University of Eindhoven, DNV (research), Phase2Phase, TecnoIution

Scale: National, the Netherlands.

NetOptimalisatie voor Grootschalige Inpassing Zon- en windstroom Middels Opslag en Software (NO-GIZMOS)

Netherlands Enterprise Agency (RVO)

📅 2022-2025

📍 Netherlands

Congestion management for Large-scale Integration of Solar and Wind Power Through Storage and Software with real demonstrator with active involvement of DSO (ENEXIS).

Consortium: ENEXIS (DSO), Technical University of Eindhoven (research), University of Applied Sciences of Groningen (research), CGI (data management), Energykansen (local energy community), iWell (Technology provider).

Scale: National, the Netherlands.

Gretha

Italian ministry of research

📅 2023

📍 Italy

Research project about developing a novel GReen Energy Technology based on fuel cells, Hydrogen And renewables.

Consortium: Universita' Politecnica delle Marche (research), ENAV (Aviation), TECNOSISTEM (engineering), UNINA (research).

Scale: National, Italy.

eNeuron

EU commission/Horizon 2020

📅 2020-2024

📍 Italy

Research project about Optimising the design and operation of local energy communities based on multi-carrier energy systems.

Consortium: ENEA(project coordination), EPRI (research), IEN, City of Bydgoszcz (local authority), ENEIDA.io(IOT), IREC (modeling), Universita' Politecnica delle Marche (research), DERlab, Iede, edp Iabeled, Technical University of Eindhoven, Sintef, Tecnalia, Universidad Politecnica de Madrid, ENEA operator, ICONS, Marinha Portuguesa, FOSS.

Scale: International, Europe.

Ricerca del sistema elettrico italiano

Italian ministry of research

📅 2020-2022

📍 Italy

Research project about the development of a methodology for optimal planning for a 'Multi-energy system Energy Community' for medium-long term'.

Consortium: ENEA, Universita' Politecnica delle Marche research.

Scale: National, Italy.

TEACHING EXPERIENCE

Date	Lecture Topic	Audience	Institution
March 2026	Marker participation of renewable producers	MSc Electrical Engineering (5LEPO)	TU Eindhoven, NL
Oct. 2025	Energy storage and their role in the power systems	MSc Electrical Engineering	TU Eindhoven, NL
March 2025	Master Lecture: How to build an optimization model	MSc Sustainable Energy Technology	TU Eindhoven, NL
March 2022	Sistemi di accumulo e loro gestione	MSc Mechanical Engineering	UNIVPM, IT
March 2022	Pyomo optimization model set-up	MSc Mechanical Engineering	UNIVPM, IT

MASTER THESIS DAILY SUPERVISION

58. Chengyuan Guan, *Optimal Sizing of Hybrid Renewable Energy Solutions for Data Centers: Case Studies of Non-Firm Grid, Reduced- and Off-Grid Scenarios*, MSc Sustainable Energy Technology, Technical University of Eindhoven, 2025.
57. Dennis Hollanders, *Graph Neural Networks for Distribution Network Reconfiguration Optimization*, Department of Industrial Engineering & Innovation Sciences Information Systems Research Group, Technical University of Eindhoven, 2025.

- S6. Stefan De Lange, *Hybrid Heat Pump Optimization for Flexibility Provision: Modeling and Simulation*, Department of Electrical Engineering, Technical University of Eindhoven, 2024.
- S5. Fabian Caipa Cure, *Adaptive Distributionally Robust Optimization for Residential Energy Management under Non-firm Capacity Contracts*, Department of Mechanical Engineering, Master's in Sustainable Energy Technology, Technical University of Eindhoven, 2024.
- S4. Francesco Panara, *Study of Hydrogen-to-Power systems: state-of-the-art of alkaline and Proton Exchange Membrane fuel cells and performance evaluation through Python modelling*, Dipartimento di Ingegneria Industriale e Scienze Matematiche, Università Politecnica delle Marche, 2023.
- S3. Filippo Onori, *Design and management of a "Battery Energy Storage System" (BESS) to provide flexibility service to the national electricity grid*, Dipartimento di Ingegneria Industriale e Scienze Matematiche, Università Politecnica delle Marche, 2023.
- S2. Francesca Mennilli, *Study of systems related to Power-to-Hydrogen: state of art of the electrolyser and its modeling using Python*, Dipartimento di Ingegneria Industriale e Scienze Matematiche, Università Politecnica delle Marche, 2022.
- S1. Luca Ciotti, *Study of systems related to Power-to-Hydrogen: state of art of the of the storage and its modeling using Python*, Dipartimento di Ingegneria Industriale e Scienze Matematiche, Università Politecnica delle Marche, 2022.

PUBLICATIONS

Journal Articles

- [12] **Lingkang Jin**, Laurens Blik, and Nikolaos G. Paterakis. "Learning Active Constraints for Bilevel Optimization: Evaluating Classification Metrics on Real-World Distribution System Data". In: *IEEE Transactions on Smart Grid* 17.2 (2026), pp. 1303–1314. DOI: 10.1109/TSG.2025.3632419.
- [11] **Lingkang Jin**, Rafael Nogueira Nakashima, Gabriele Comodi, and Henrik Lund Frandsen. "Alkaline electrolysis for green hydrogen production: A novel, simple model for thermo-electrochemical coupled system analysis". In: *Applied Thermal Engineering* 262 (2025), p. 125154. ISSN: 1359-4311. DOI: <https://doi.org/10.1016/j.applthermaleng.2024.125154>. URL: <https://www.sciencedirect.com/science/article/pii/S1359431124028229>.
- [10] **Lingkang Jin**, Mosè Rossi, Andrea Monforti Ferrario, Francesca Mennilli, and Gabriele Comodi. "Designing hybrid energy storage systems for steady green hydrogen production in residential areas: A GIS-based framework". In: *Applied Energy* 389 (2025), p. 125765. ISSN: 0306-2619. DOI: <https://doi.org/10.1016/j.apenergy.2025.125765>. URL: <https://www.sciencedirect.com/science/article/pii/S0306261925004957>.
- [9] **Lingkang Jin**, Milad Kazemi, Gabriele Comodi, and Christina Papadimitriou. "Assessing battery degradation as a key performance indicator for multi-objective optimization of multi-carrier energy systems". In: *Applied Energy* 361 (2024), p. 122925. ISSN: 0306-2619. DOI: <https://doi.org/10.1016/j.apenergy.2024.122925>. URL: <https://www.sciencedirect.com/science/article/pii/S0306261924003088>.
- [8] Francesca Mennilli, **Lingkang Jin**, Mosè Rossi, and Gabriele Comodi. "Assessment of a NaOH-based alkaline electrolyser's performance: System modelling and operating parameters optimisation". In: *International Journal of Hydrogen Energy* 85 (2024), pp. 625–634. ISSN: 0360-3199. DOI: <https://doi.org/10.1016/j.ijhydene.2024.08.175>. URL: <https://www.sciencedirect.com/science/article/pii/S0360319924033342>.
- [7] Andrea Pizzuti, **Lingkang Jin**, Mosè Rossi, Fabrizio Marinelli, and Gabriele Comodi. "A novel approach for multi-stage investment decisions and dynamic variations in medium-term energy planning for multi-energy carriers community". In: *Applied Energy* 353 (2024), p. 122177. ISSN: 0306-2619. DOI: <https://doi.org/10.1016/j.apenergy.2023.122177>. URL: <https://www.sciencedirect.com/science/article/pii/S0306261923015416>.
- [6] Mosè Rossi, **Lingkang Jin**, Andrea Monforti Ferrario, Marialaura Di Somma, Amedeo Buonanno, Christina Papadimitriou, Andrei Morch, Giorgio Graditi, and Gabriele Comodi. "Energy Hub and Micro-Energy Hub Architecture in Integrated Local Energy Communities: Enabling Technologies and Energy Planning Tools". In: *Energies* 17.19 (2024). ISSN: 1996-1073. DOI: 10.3390/en17194813. URL: <https://www.mdpi.com/1996-1073/17/19/4813>.
- [5] **Lingkang Jin**, Mosè Rossi, Lucio Ciabattani, Marialaura Di Somma, Giorgio Graditi, and Gabriele Comodi. "Environmental constrained medium-term energy planning: The case study of an Italian university campus as a multi-carrier local energy community". In: *Energy Conversion and Management* 278 (2023), p. 116701. ISSN: 0196-8904. DOI: <https://doi.org/10.1016/j.enconman.2023.116701>. URL: <https://www.sciencedirect.com/science/article/pii/S019689042300047X>.
- [4] **Lingkang Jin**, Mosè Rossi, Andrea Monforti Ferrario, Jacopo Carlo Alberizzi, Massimiliano Renzi, and Gabriele Comodi. "Integration of battery and hydrogen energy storage systems with small-scale hydropower plants in off-grid local energy communities". In: *Energy Conversion and Management* 286 (2023), p. 117019. ISSN: 0196-8904. DOI: <https://doi.org/10.1016/j.enconman.2023.117019>. URL: <https://www.sciencedirect.com/science/article/pii/S0196890423003655>.
- [3] Enrico Marchegiani, Francesco Ferracuti, Andrea Monteriù, **Lingkang Jin**, Mosè Rossi, Gabriele Comodi, and Lucio Ciabattani. "Li-ion battery aging model robustness: An analysis using univariate and multivariate techniques". In: *Journal of Energy Storage* 72 (2023), p. 108591. ISSN: 2352-152X. DOI: <https://doi.org/10.1016/j.est.2023.108591>. URL: <https://www.sciencedirect.com/science/article/pii/S2352152X23019886>.
- [2] **Lingkang Jin**, Andrea Monforti Ferrario, Viviana Cigolotti, and Gabriele Comodi. "Evaluation of the impact of green hydrogen blending scenarios in the Italian gas network: Optimal design and dynamic simulation of operation strategies". In: *Renewable and Sustainable Energy Transition* 2 (2022), p. 100022. ISSN: 2667-095X. DOI: <https://doi.org/10.1016/j.rset.2022.100022>. URL: <https://www.sciencedirect.com/science/article/pii/S2667095X2200006X>.
- [1] Anam Nadeem, Mosè Rossi, Erica Corradi, **Lingkang Jin**, Gabriele Comodi, and Nadeem Ahmed Sheikh. "Energy-Environmental Planning of Electric Vehicles (EVs): A Case Study of the National Energy System of Pakistan". In: *Energies* 15.9 (2022). ISSN: 1996-1073. DOI: 10.3390/en15093054. URL: <https://www.mdpi.com/1996-1073/15/9/3054>.

- [13] **Lingkang Jin**, Sen Zhan, Samuel Cudjoe, and Nikolaos G. Paterakis. "Empowering Low-Voltage Grids: Real-World Implementation of Home Batteries for Effective Congestion Management". In: *2025 IEEE Kiel PowerTech*. 2025, pp. 1–6. DOI: 10.1109/PowerTech59965.2025.11180463.
- [12] **Lingkang Jin**, Sen Zhan, Haoyang Zhang, and Nikolaos G. Paterakis. "Multi-Objective Congestion Management in Low-Voltage Grids: Coordinating End-Users and System Operators for Real-World Implementation". In: *2025 IEEE PES Innovative Smart Grid Technologies Conference Europe (ISGT Europe)*. 2025, pp. 1–5. DOI: 10.1109/ISGTEurope64741.2025.11305401.
- [11] Xin Li, Jeroen J. Markus, Stefan De Lange, **Lingkang Jin**, Koen Kok, and Nikolaos G. Paterakis. "Hybrid Heat Pump Flexibility Allocation: Quantified Thermal Comfort-Based Congestion Management". In: *2025 IEEE International Conference on Environment and Electrical Engineering and 2025 IEEE Industrial and Commercial Power Systems Europe (EEEIC / ICPS Europe)*. 2025, pp. 1–6. DOI: 10.1109/EEEIC/ICPSEurope64998.2025.11169153.
- [10] Haoyang Zhang, **Lingkang Jin**, Koen Kok, and Nikolaos G. Paterakis. "Surrogate Model-Based Reinforcement Learning for Bidding Strategies in Local Flexibility Markets". In: *2025 IEEE International Conference on Environment and Electrical Engineering and 2025 IEEE Industrial and Commercial Power Systems Europe (EEEIC / ICPS Europe)*. 2025, pp. 1–6. DOI: 10.1109/EEEIC/ICPSEurope64998.2025.11169071.
- [9] **Lingkang Jin**, Xin Li, Stefan De Lange, Han Slootweg, and Nikolaos G. Paterakis. "Response Allocation of Domestic Hybrid Heat Pumps Flexibility for Congestion Management". In: *2024 IEEE PES Innovative Smart Grid Technologies Europe (ISGT EUROPE)*. 2024, pp. 1–5. DOI: 10.1109/ISGTEUROPE62998.2024.10863630.
- [8] Francesca Mennilli, **Lingkang Jin**, Mosè Rossi, and Gabriele Comodi. "A Fitting Process for the Optimal Modelling of an Anion Exchange Membrane (AEM) Electrolyser". In: *Proceedings of the ASME Turbo Expo 2024: Power for Land, Sea, and Air*. Vol. 2: Ceramics and Ceramic Composites; Coal, Biomass, Hydrogen, and Alternative Fuels. June 2024, V002T03A017.
- [7] Marialaura Di Somma, **Lingkang Jin**, Nicola Bianco, and Christina Papadimitriou. "Advancing Multi-Energy Hub Design: an Integrated Approach for Optimizing Residential Clusters in High RES Penetration Scenarios". In: *2024 International Conference on Smart Energy Systems and Technologies (SEST)*. 2024, pp. 1–6. DOI: 10.1109/SEST61601.2024.10694519.
- [6] J. C. Alberizzi, M. A. Pérez Estevez, M. Renzi, **L. Jin**, M. Rossi, and A. Alberizzi. "Optimal Management of a Hydro – Wind Energy System with Hydrogen Storage". In: *2023 12th International Conference on Power Science and Engineering (ICPSE)*. 2023, pp. 46–50. DOI: 10.1109/ICPSE59506.2023.10329307.
- [5] A Bechmann, T Barlas, H L Frandsen, L Jin, and R N Nakashima. "The Hydrogen Wind Turbine: Design of a wind turbine optimised for hydrogen production". In: vol. 2507. 1. IOP Publishing, May 2023, p. 012010. DOI: 10.1088/1742-6596/2507/1/012010. URL: <https://dx.doi.org/10.1088/1742-6596/2507/1/012010>.
- [4] Marialaura Di Somma, Amedeo Buonanno, Martina Caliano, **Lingkang Jin**, Mosè Rossi, Giorgio Graditi, and Gabriele Comodi. "Stochastic energy management for the Italian UNIVPM campus as a multi-carrier energy hub participating in the day-ahead market". In: *IEEE EUROCON 2023 - 20th International Conference on Smart Technologies*. 2023, pp. 251–256. DOI: 10.1109/EUROCON56442.2023.10199000.
- [3] **Lingkang Jin**, Rafael Nogueira Nakashima, Henrik Lund Frandsen, and Gabriele Comodi. "Alkaline Electrolysis for Green Hydrogen Production: Techno-Economic Analysis of Temperature Influence and Control". In: *Proceedings of the 36th International Conference on Efficiency, Cost, Optimization, Simulation and Environmental Impact of Energy Systems (ECOS)*. 2023. DOI: 10.52202/069564-0082. URL: <https://doi.org/10.52202/069564-0082>.
- [2] **Lingkang Jin**, Mosè Rossi, Flavio Caresana, Leonardo Pelagalli, and Gabriele Comodi. "Metal hydrides in hydrogen storage: optimization of dynamic control strategies". In: vol. 2648. 1. IOP Publishing, Dec. 2023, p. 012056. DOI: 10.1088/1742-6596/2648/1/012056. URL: <https://dx.doi.org/10.1088/1742-6596/2648/1/012056>.
- [1] Francesca Mennilli, **Lingkang Jin**, Mosè Rossi, Alice Mugnini, and Gabriele Comodi. "Energy analysis of a hydrogen integrated system in the residential sector". In: vol. 2648. 1. IOP Publishing, Dec. 2023, p. 012057. DOI: 10.1088/1742-6596/2648/1/012057. URL: <https://dx.doi.org/10.1088/1742-6596/2648/1/012057>.